CPPC 21/L

DIGITAL SIGNAL PROCESSING

ACTIVITY No.3

Histogram and Grayscale

Submitted by:

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BS CpE 3B

Submitted to:

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Subject Professor

I. Objectives

- To study the distribution of pixel intensity values using a histogram.
- To convert a colored image into a grayscale image.
- To understand how histogram equalization improves image contrast.
- To compare the histogram of the original image and the grayscale image.
- To observe how MATLAB can be used for basic image processing.

II. Materials

- MATLAB
- Laptop
- Sample Image

III. Procedures

1. Open MATLAB

- Launch the MATLAB application.
- Click New Script to create a new script file.
- Upload or place the sample image in the MATLAB working folder.
- Type the command below in the script:

```
amber = imread("amber.jpg");
```

- After preparing the script, proceed to the Command Window.

2. Load the Image

- In the Command Window, type the command:

```
amber = imread('amber.jpg')
```

- This command loads the image into MATLAB.
- The image will be stored in the variable named jake.

3. Display the Original Image

- To show the original colored image, type:

  imshow(amber)

- This displays the image before any changes or processing are applied.

4. Show the Histogram of the Original Image

- To view the histogram of the original image, type:

  imhist(amber)

- The histogram shows the distribution of pixel intensity values from the image.

5. Apply Histogram Equalization

- To improve the contrast of the original image, type:

  equalized = histeq(amber);

- This process adjusts the image intensity values to make the details more visible.

6. Display the Equalized Image

- To show the enhanced image, type:

  imshow(equalized)

- This displays the result after applying histogram equalization.

7. Convert the Image to Grayscale

- To convert the colored image into grayscale, type:

  gray = rgb2gray(amber)

- This removes the color information and changes the image into shades of gray.

8. Display the Grayscale Image

- To show the grayscale version of the image, type:

  imshow(gray)

- This displays the image using only black, white, and gray intensity levels.

9. Show the Histogram of the Grayscale Image

- To view the histogram of the grayscale image, type:

  imhist(gray)

- This shows how the pixel intensities are distributed in the grayscale image.

10. Apply Histogram Equalization to the Grayscale Image

- To enhance the grayscale image, type:

  equalized_gray = histeq(gray);

- This improves the contrast by spreading the grayscale intensity values more evenly.

11. Display the Equalized Grayscale Image

- To show the enhanced grayscale output, type:

  imshow(equalized_gray)

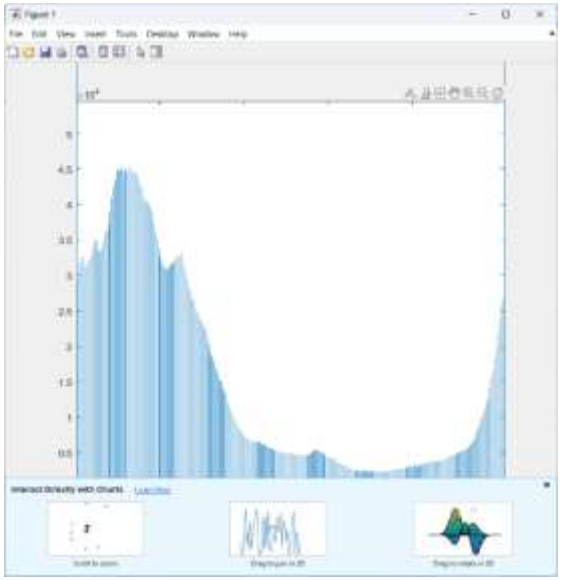
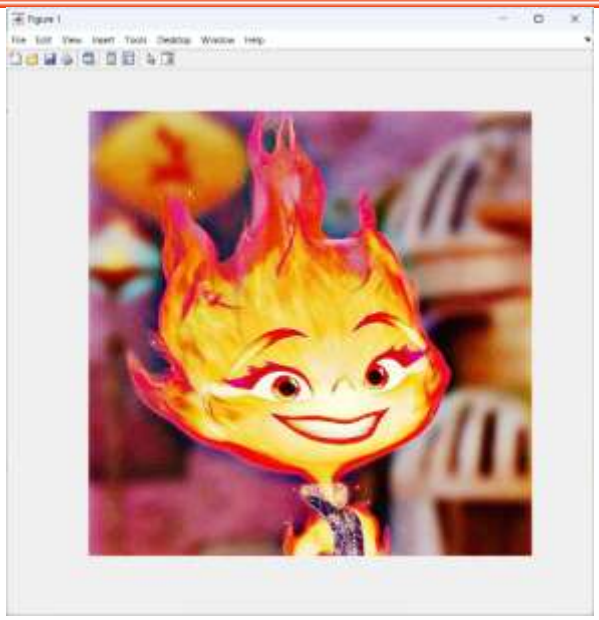
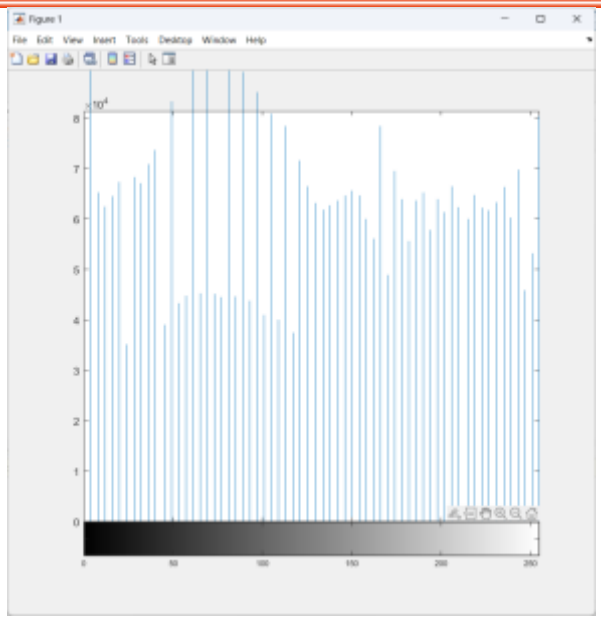
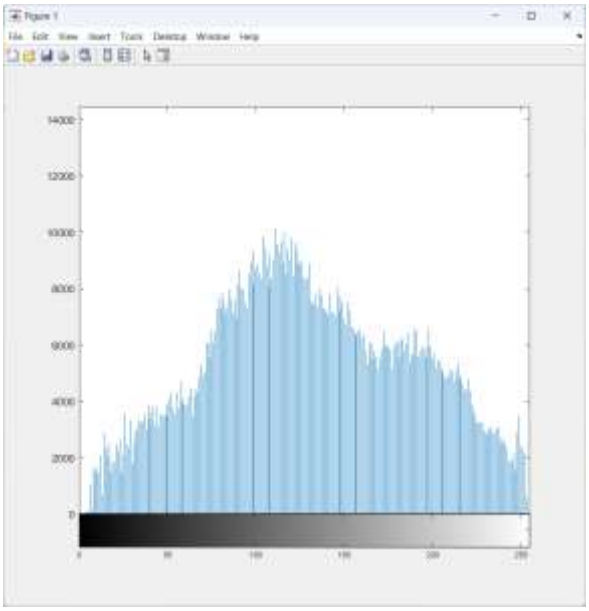
- This displays the grayscale image after histogram equalization.

12. Compare the Results

- Compare the original image with the grayscale image.
- Compare the histogram of the original image with the histogram of the grayscale image.

- Observe how histogram equalization affects the brightness, contrast, and visibility of image details.
- Check how the grayscale image changes after applying histogram equalization.

IV. Conclusion

The activity showed how MATLAB can be used to analyze and improve images using histograms and grayscale conversion. By displaying the histogram of the original image, the distribution of pixel intensities was observed, and histogram equalization helped enhance the image contrast. After converting the image to grayscale, the histogram was also examined and equalization was applied again to improve the visibility of details. Overall, this activity helped demonstrate how image processing techniques can modify and enhance visual information using MATLAB.

V. Documentation